

Jaemin Kim

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RESEARCH INTEREST

Aiming to improve the controllability of advance generative models, including Diffusion Models and Large Language Models (LLMs). Recently contributed to this goal by exploring efficient guidance methods, such as training-free approaches and general-purpose strategies.

EDUCATION

Korea Advanced Institute of Science and Technology (KAIST) <i>Ph.D. in Artificial Intelligence</i>	Mar 2025 - Now <i>Advisor: Prof. Jong Chul Ye</i>
Korea Advanced Institute of Science and Technology (KAIST) <i>M.S. in Artificial Intelligence</i>	Mar 2023 - Feb 2025 <i>Advisor: Prof. Jong Chul Ye</i>
Korea Advanced Institute of Science and Technology (KAIST) <i>B.S. in Bio and Brain Engineering, Electrical Engineering (Double Major)</i>	Mar 2016 - Feb 2023 <i>GPA: 3.96/4.3 (Magna Cum Laude)</i>
Karlsruhe Institute of Technology (KIT) <i>Electrical Engineering and Information Technology (Exchange Student)</i>	Sep 2019 - Feb 2020 <i>Karlsruhe, Germany</i>

PUBLICATIONS *: Equal Contribution

[C6] <u>Universal Reasoner: A Single, Composable Plug-and-Play Reasoner for Frozen LLMs</u> Jaemin Kim*, Hangeol Chang*, Hyunmin Hwang*, Choonghan Kim, Jong Chul Ye	ICML 2026
[C5] <u>Training-Free Reward-Guided Image Editing via Trajectory Optimal Control</u> Jinho Chang*, Jaemin Kim*, Jong Chul Ye	ICLR 2026
[C4] <u>Free²Guide: Training-Free Text-to-Video Alignment using Image LVLM</u> Jaemin Kim, Bryan Sangwoo Kim, Jong Chul Ye	ICCV 2025
[C3] <u>Optical-Flow Guided Prompt Optimization for Coherent Video Generation</u> Hyelin Nam*, Jaemin Kim*, Dohun Lee, Jong Chul Ye	CVPR 2025
[C2] <u>Derivative-Free Diffusion Manifold-Constrained Gradient for Unified XAI</u> Won Jun Kim*, Hyungjin Chung*, Jaemin Kim*, Sangmin Lee, Byeongsu Sim, Jong Chul Ye	CVPR 2025
[C1] <u>Generalized Consistency Trajectory Models for Image Manipulation</u> Beomsu Kim*, Jaemin Kim*, Jeongsol Kim, Jong Chul Ye	ICLR 2025

PRE-PRINTS *: Equal Contribution

[P5] <u>AgentPSO: Evolving Agent Reasoning Skill via Multi-agent Particle Swarm Optimization</u> Hyunmin Hwang*, Jaemin Kim*, Choonghan Kim, Hangeol Chang, Jong Chul Ye	Preprint 2026
[P4] <u>Adaptive Guidance for Retrieval-Augmented Masked Diffusion Models</u> Jaemin Kim, Jong Chul Ye	Preprint 2026

[P3] <u>DMD-augmented Unpaired Neural Schrödinger Bridge for Ultra-Low Field MRI Enhancement</u>	<i>Preprint 2026</i>
Youngmin Kim*, Jaeyun Shin*, Jeongchan Kim*, Taehoon Lee, Jaemin Kim , Peter Hsu, Jelle Veraart, Jong Chul Ye	
[P2] <u>Dementia-R1: Reinforced Pretraining and Reasoning from Unstructured Clinical Notes for Real-World Dementia Prognosis</u>	<i>Preprint 2026</i>
Choonghan Kim*, Hyunmin Hwang*, Hangeol Chang*, Jaemin Kim* , Jinse Park, Jae-Sung Lim, Jong Chul Ye	
[P1] <u>HiCBridge: Resolution Enhancement of Hi-C Data Using Direct Diffusion Bridge</u>	<i>Preprint 2024</i>
Jaemin Kim* , Jong Chul Ye	

EXPERIENCES

Research Intern <i>Sony Group</i>	May 2026 - Oct 2026 <i>Tokyo Osaka, Japan</i>
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PATENTS

Method and Apparatus for Approximating Gradient of Artificial Neural Network Model • (South Korea) Patent No.10-2025-0145337	2025
A Single, Composable Plug-and-play Reasoner for Frozen LLMs • (South Korea) Patent No.10-2025-0132734	2025

HONORS AND AWARDS

32nd Samsung Humantech Paper Award <i>Universal Reasoner: A Single, Composable Plug-and-Play Reasoner for Frozen LLMs</i>	Silver Prize <i>Track: Signal Processing</i>
Dongwon-KAIST scholarship <i>Full scholarship for tuition and stipend</i>	Mar 2023 - Feb 2025
Baden-Württemberg Stipendium <i>Scholarship for international students to study at a university in Baden-Württemberg</i>	Sep 2019 - Feb 2020
South Korea National Science & Technology Scholarship <i>Scholarship for attracting outstanding talents</i>	Mar 2018 - Mar 2020

ACADEMIC ACTIVITIES

Reviewer <i>ICLR 2026, ICML 2026</i>	
Persident, Student Council <i>KAIST AI</i>	May 2025 - Mar 2026
Teaching Assistant <i>KAIST - AI618: Generative Models and Unsupervised Learning</i>	Mar 2025 - Jun 2025

REFERENCES

Cristian Simon <i>u6562565@anu.edu.au</i>	Mentor at Sony <i>AI Researcher, Sony Group</i>
Jong Chul Ye <i>jong.ye@kaist.ac.kr</i>	M.S., Ph.D. Advisor <i>Endowed Chair Professor, Kim Jaechul Graduate School of Artificial Intelligence, KAIST</i>